

Decision Memo

Bottom Line: 22% of customers (“Champions”) generate 65% of revenue and should be protected. The clearest opportunity for retention spend is the “At Risk” segment: 10.5% of customers, representing roughly \$742K in historical revenue, who have gone 142 days without a purchase despite a strong buying history. Recommend prioritizing a win-back campaign for “At Risk” customers first, light-touch loyalty perks for “Champions” second, and no dedicated retention spend on “Hibernating” customers, whose expected return is low.

Business Question: “Which customer segments are most valuable, and which are at risk of churning and how should the retention budget be allocated across them?”

Data Source: Online Retail II, Kaggle, (Chen, London South Bank University)

Background: The data obtained contains over 540,000 rows of online shopping data from a UK-based retailer between 01/12/2010 and 09/12/2011. The relevant variables include Customer ID (customer identifier), Invoice (order/transaction identifier), InvoiceDate (date/time of purchase), and Quantity × Price (combined into TotalPrice). Records with missing customer IDs, cancelled orders, and returns/negative quantities were excluded as they aren’t full purchases.

Methodology: RFM analysis (Recency, Frequency, Monetary) was run on 12 months of transaction-level order data (4,338 customers, ~398K line items after cleaning). Each customer was scored 1-5 on each dimension and grouped into six standard segments: “Champions”, “Loyal Customers”, “At Risk”, “Needs Attention”, “Hibernating”, and “New Customers”.

Findings:

- The “Champions” segment drives 65.2% of the revenue, but make up 22.2% of customers
- “At Risk” customers haven’t purchased in 142 days on average despite strong spending
- The “Needs Attention” segment is the largest percentage of the customer base at 23.5%, but only drive 7.0% of the revenues
- “Hibernating” customers (19% of the customers) generate only 2.1% of the revenue
- “New Customers” (7.4% of the customers) are recent and need more data to make conclusive statements regarding retention and revenue

Recommendations:

- “At Risk”: launch a targeted win-back campaign (personalized discount or outreach)
- “Champions”: protect with light loyalty perks, rewards, and discounts to retain them
- “Needs Attention” & “New Customers”: low-cost outreach like cold emailing to increase retention rates and keep them engaged
- “Hibernating”: minimal historical value therefore reactivation spend unlikely to pay back

Caveats: Since the data is across mainly the UK/Europe and only covers up to 1 year worth of data, these findings and coding decisions pertain only to this dataset. Recency is measured against the last date in the dataset, not the current date.